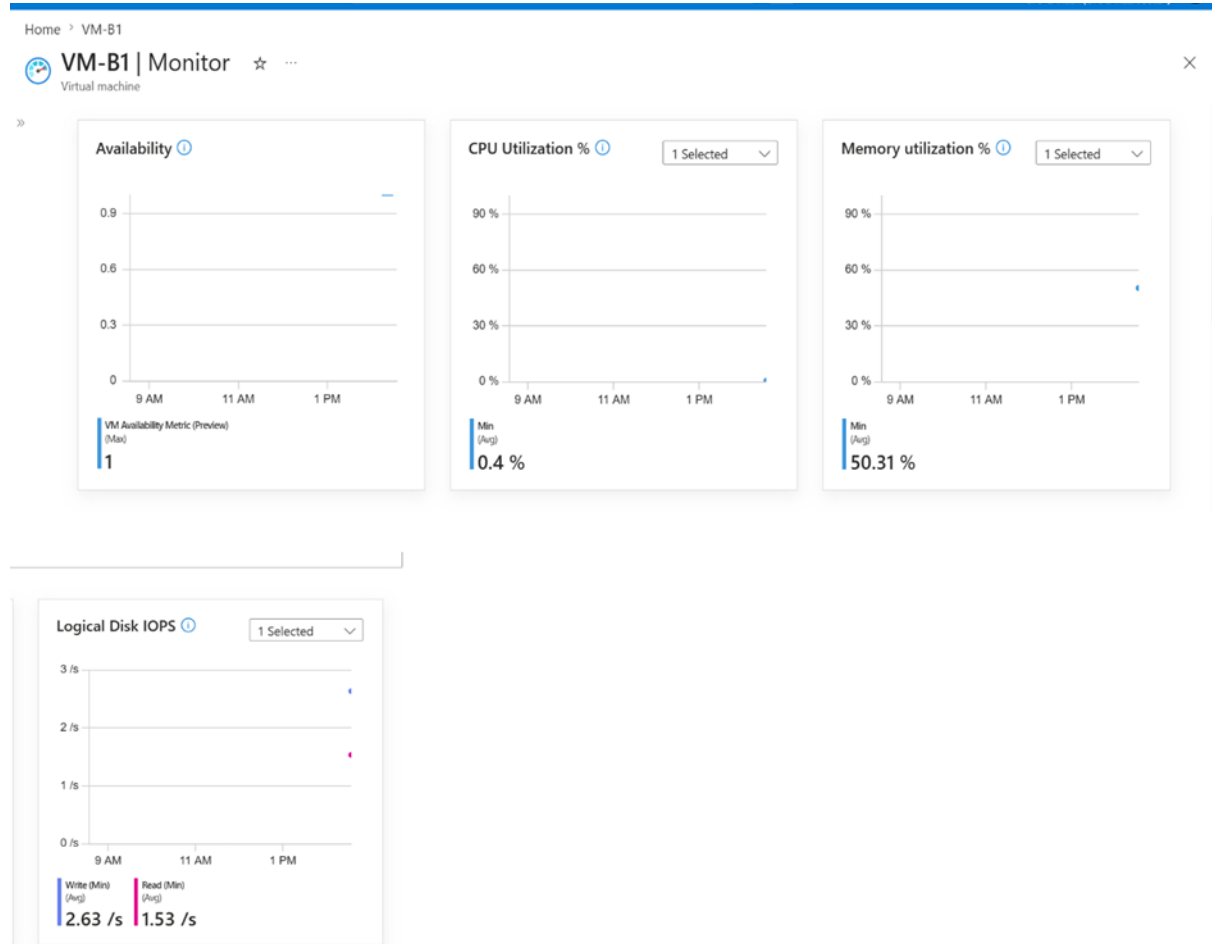


WEEK 7 PRACTICAL

Task 1 enable azure monitoring



enabled Microsoft azure monitoring for the virtual machine **VM-B1** to collect and display important performance metrics such as:

- CPU utilization
- Memory usage
- VM availability
- Logical disk IOPS

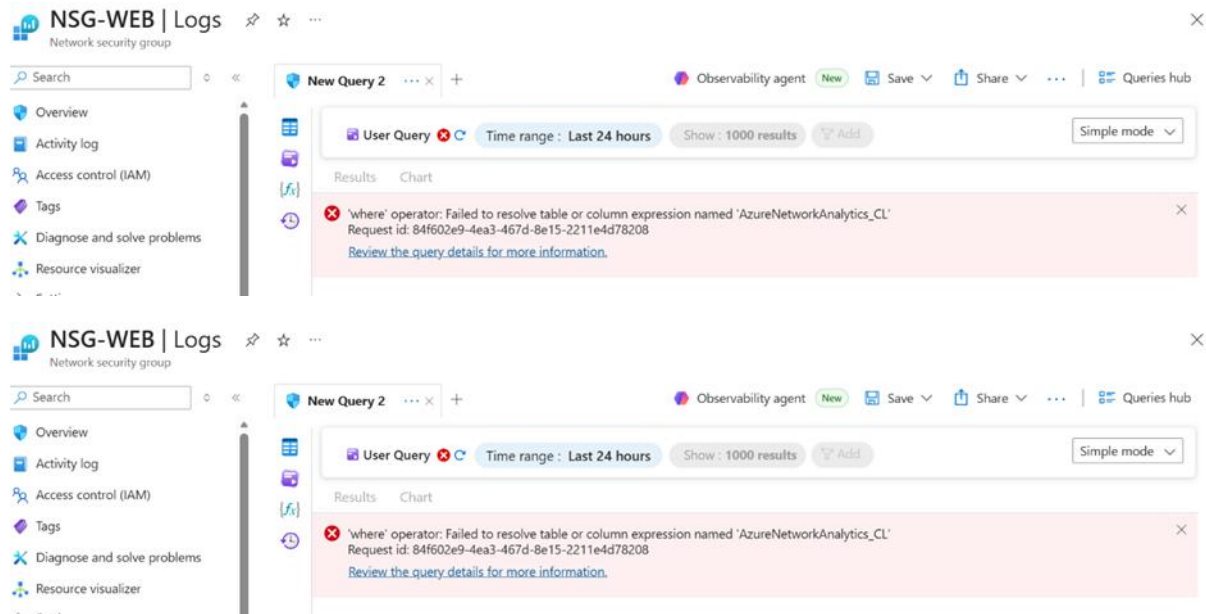
The monitoring data collected by Azure Monitor is stored in **Azure Monitor Metrics** and can also be integrated with a **Log Analytics Workspace** for:

- Long-term data storage
- Advanced monitoring and analysis
- Creating alerts and notifications

- Troubleshooting performance issues

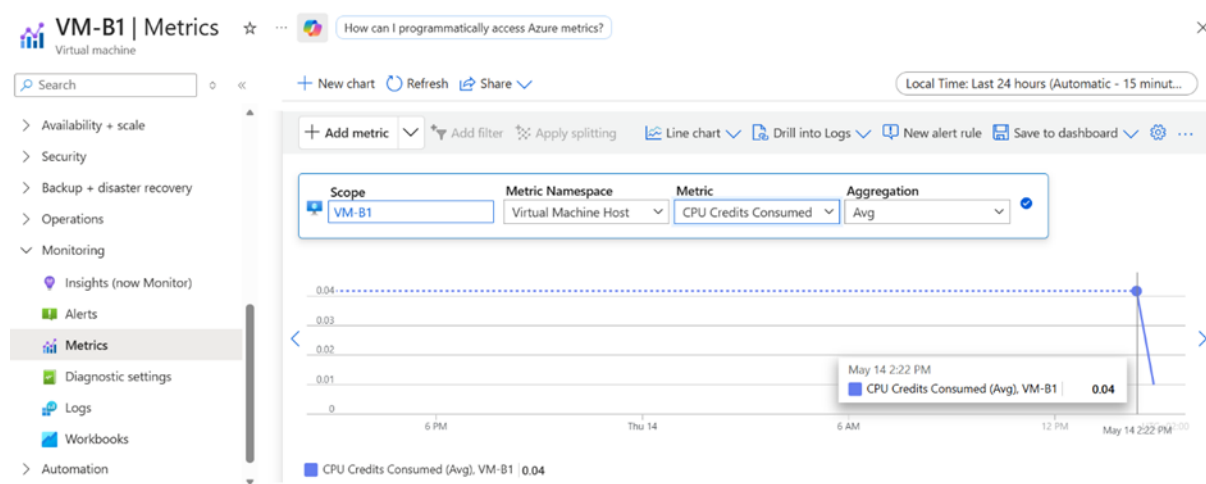
This setup helps administrators monitor the health and performance of the virtual machine in real time and respond quickly to any potential issues.

Task 1.2 enable monitoring on network resources



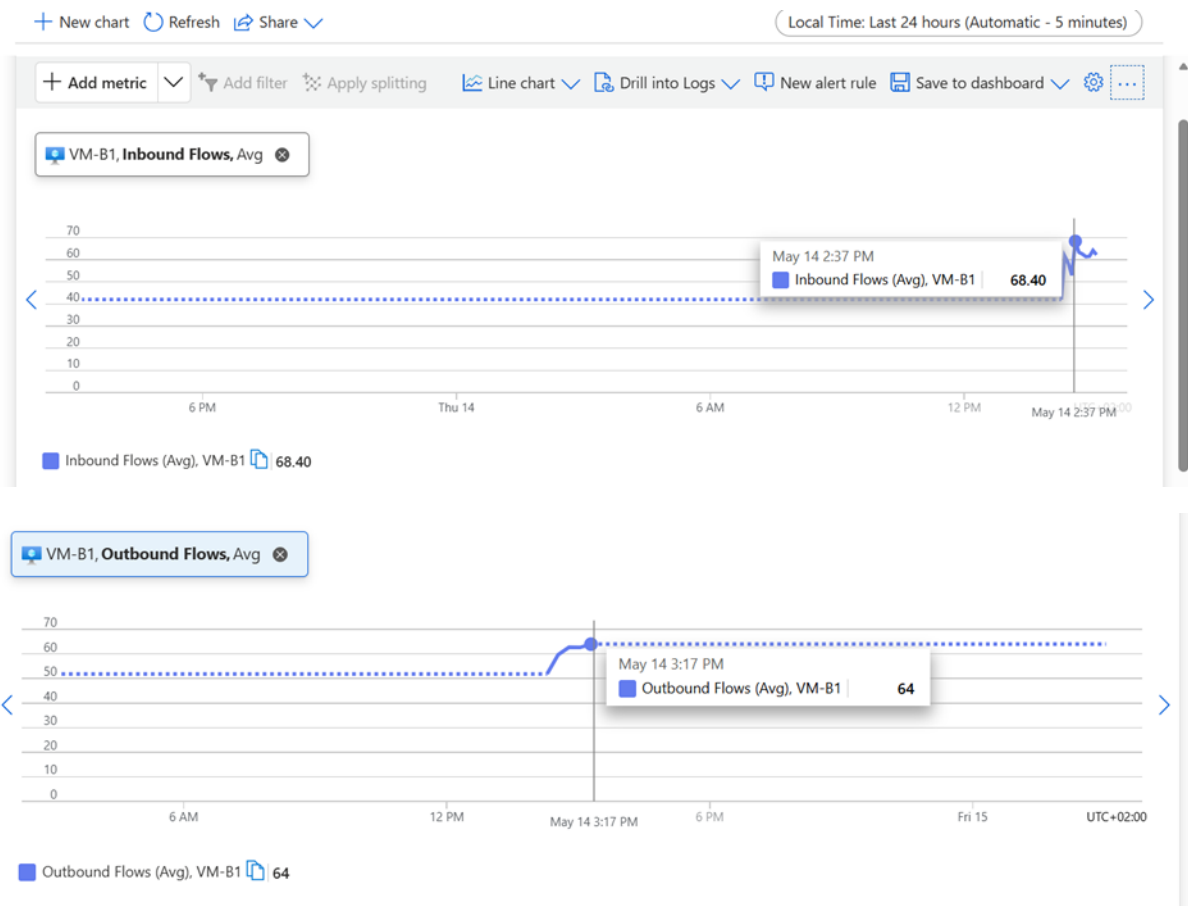
Task 2

Task 2.1 CPU metrics



The CPU metric indicates that the virtual machine is using very few CPU credits, with usage remaining at approximately 0.04 on average. This suggests the VM is operating under a low workload, which is expected for a system running minimal applications or background services.

Task 2.2



Task 3.2 configure an alert rule

The screenshot shows the 'VM-B1 | Alerts' page in Azure Monitor. The 'Select alert rules' section is active, showing a rule 'Percentage CPU is greater than' with a threshold of 80%. Below this, a summary of alerts is displayed: Total alerts 0, Critical 0, Error 0, Warning 0, Informational 0, and Verbose 0. The page also includes a sidebar with navigation links for Security, Backup + disaster recovery, Operations, Monitoring, Insights (now Monitor), and Alerts.

Reflection Question	Your Response
Why is monitoring important in a cloud environment? How does it differ from monitoring on-premises infrastructure?	Monitoring is important in cloud environments to ensure resources perform well, remain secure, and stay cost-effective. Unlike on-premises infrastructure, cloud monitoring focuses on virtual resources, scaling, and usage-based costs.

What is the difference between a metric and a log in Azure Monitor? When would you use each?	Metrics are numerical performance values such as CPU usage or request count and are used for real-time monitoring and alerts. Logs are detailed event records used for troubleshooting and analysing issues. Metrics show what happened, while logs help explain why it happened.
If your VM showed consistently high CPU usage (above 90%) every day at 2am, what steps would you take to investigate?	I would first confirm the pattern using Azure Monitor metrics, then check for scheduled tasks, backups, or updates running at 2am. I would also use Log Analytics to identify the process causing the high CPU usage and then optimize, reschedule, or scale the VM if needed.
How could monitoring data be used to make cost-optimisation decisions in Azure?	Monitoring data helps identify underused or idle resources that can be resized or removed to reduce costs. It can also improve autoscaling settings and help decide when to use Reserved Instances or Savings Plans for discounts.